

Week-2

April 23, 2026

0.1 Import necessary libraries

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

0.2 Task 1: Read the Pantheon+SH0ES data table

```
[2]: # .dat files in astronomy are typically space-separated or tab-separated.
# We use '\s+' to handle any amount of whitespace between columns.
file_path = 'data/Pantheon+SH0ES.dat'

try:
    df = pd.read_csv(file_path, sep=r'\s+')
    print("Dataset loaded successfully.")
    #print the head and tail of the data file
    display(df.head())
    display(df.tail())
except FileNotFoundError:
    print(f"Error: The file {file_path} was not found. Please ensure it is in_
    ↳the same directory.")
```

Dataset loaded successfully.

	CID	IDSURVEY	zHD	zHDERR	zCMB	zCMBERR	zHEL	\
0	2011fe	51	0.00122	0.00084	0.00122	0.00002	0.00082	
1	2011fe	56	0.00122	0.00084	0.00122	0.00002	0.00082	
2	2012cg	51	0.00256	0.00084	0.00256	0.00002	0.00144	
3	2012cg	56	0.00256	0.00084	0.00256	0.00002	0.00144	
4	1994DRichmond	50	0.00299	0.00084	0.00299	0.00004	0.00187	

	zHELERR	m_b_corr	m_b_corr_err_DIAG	...	PKMJDERR	NDOF	FITCHI2	\
0	0.00002	9.74571	1.516210	...	0.1071	36	26.8859	
1	0.00002	9.80286	1.517230	...	0.0579	101	88.3064	
2	0.00002	11.47030	0.781906	...	0.0278	165	233.5000	
3	0.00002	11.49190	0.798612	...	0.0667	55	100.1220	
4	0.00004	11.52270	0.880798	...	0.0522	146	109.8390	

	FITPROB	m_b_corr_err_RAW	m_b_corr_err_VPEC	biasCor_m_b	biasCorErr_m_b	\
0	0.864470	0.0991	1.4960	0.0381	0.005	
1	0.812220	0.0971	1.4960	-0.0252	0.003	
2	0.000358	0.0399	0.7134	0.0545	0.019	
3	0.000193	0.0931	0.7134	0.0622	0.028	
4	0.988740	0.0567	0.6110	0.0650	0.009	

	biasCor_m_b_COVSCALE	biasCor_m_b_COVADD
0	1.0	0.003
1	1.0	0.004
2	1.0	0.036
3	1.0	0.040
4	1.0	0.006

[5 rows x 47 columns]

	CID	IDSURVEY	zHD	zHDERR	zCMB	zCMBERR	zHEL	\
1696	rutledge	106	1.61505	0.00545	1.61499	0.005	1.61399	
1697	geta	106	1.69706	0.04006	1.69702	0.040	1.70000	
1698	stone	106	1.80119	0.02014	1.80111	0.020	1.80000	
1699	wilson	106	1.91165	0.00263	1.91160	0.001	1.91401	
1700	colfax	106	2.26137	0.02018	2.26130	0.020	2.26000	

	zHELERR	m_b_corr	m_b_corr_err_DIAG	...	PKMJDERR	NDOF	FITCHI2	\
1696	0.005	25.9065	0.331927	...	1.7256	8	2.92493	
1697	0.040	26.0333	0.379521	...	1.8973	12	9.15387	
1698	0.020	26.2335	0.280685	...	1.1631	11	11.69320	
1699	0.001	26.1703	0.357624	...	1.4902	9	5.54378	
1700	0.020	26.9298	0.280011	...	5.6828	13	10.19320	

	FITPROB	m_b_corr_err_RAW	m_b_corr_err_VPEC	biasCor_m_b	\
1696	0.93898	0.2230	0.0106	-0.0203	
1697	0.68973	0.3209	0.0748	0.2680	
1698	0.38714	0.2370	0.0358	-0.0968	
1699	0.78457	0.3168	0.0044	-0.3835	
1700	0.67807	0.4455	0.0297	-0.5356	

	biasCorErr_m_b	biasCor_m_b_COVSCALE	biasCor_m_b_COVADD
1696	0.005	1.000	0.002
1697	0.005	0.633	0.000
1698	0.004	0.641	0.000
1699	0.013	0.604	0.000
1700	0.009	0.164	0.000

[5 rows x 47 columns]

0.3 Task 2 - Confirm required columns exist

```
[3]: required_columns = ['zCMB', 'zHEL', 'm_b_corr', 'CEPH_DIST', 'IS_CALIBRATOR']
missing_cols = [col for col in required_columns if col not in df.columns]

if not missing_cols:
    print("Verification: All required columns exist in the dataset.")
else:
    print(f"Error: Missing columns: {missing_cols}")
```

Verification: All required columns exist in the dataset.

0.4 Task 3 - Count the total number of supernovae

```
[4]: total_observations = len(df)
print(f"Total number of supernova observations (rows): {total_observations}")
```

Total number of supernova observations (rows): 1701

0.5 Task 4 - Identify the subset of Cepheid calibrators

```
[5]: # The IS_CALIBRATOR column is a flag where 1 typically means Cepheid calibrators
calibrators = df[df['IS_CALIBRATOR'] == 1]
print(f"Number of Cepheid calibrator objects: {len(calibrators)}")
```

Number of Cepheid calibrator objects: 77

0.6 Task 5 - Inspect whether relevant columns contain finite values

```
[6]: print("Data Validation (Checking for NaNs or Infinite values):")
for col in required_columns:
    nan_count = df[col].isna().sum()
    inf_count = np.isinf(df[col]).sum()

    print(f" - Column '{col}': {nan_count} NaNs, {inf_count} Infs")
```

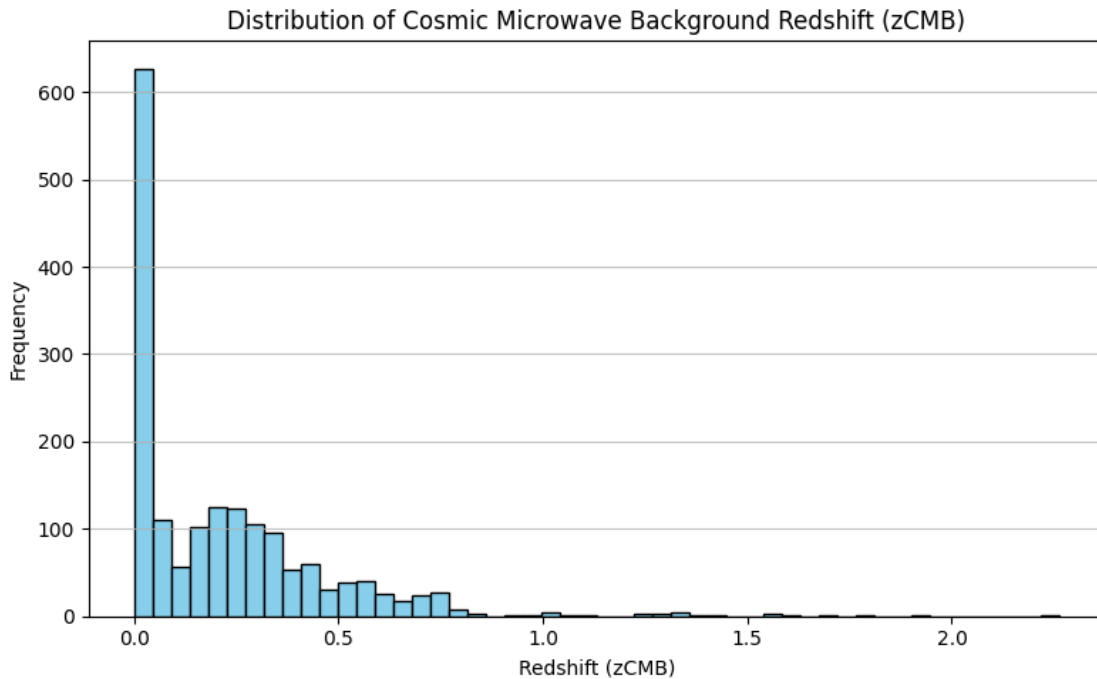
Data Validation (Checking for NaNs or Infinite values):

- Column 'zCMB': 0 NaNs, 0 Infs
- Column 'zHEL': 0 NaNs, 0 Infs
- Column 'm_b_corr': 0 NaNs, 0 Infs
- Column 'CEPH_DIST': 0 NaNs, 0 Infs
- Column 'IS_CALIBRATOR': 0 NaNs, 0 Infs

0.7 Task 6

0.7.1 Part 1 - Produce a Histogram of zCMB

```
[7]: plt.figure(figsize=(8, 5))
plt.hist(df['zCMB'], bins=50, color='skyblue', edgecolor='black')
plt.title('Distribution of Cosmic Microwave Background Redshift (zCMB)')
plt.xlabel('Redshift (zCMB)')
plt.ylabel('Frequency')
plt.grid(axis='y', alpha=0.75)
plt.tight_layout()
plt.show()
```



0.7.2 Part 2 - Produce a Scatter plot (Hubble Diagram)

```
[8]: plt.figure(figsize=(10, 6))

# Plot the standard sample (IS_CALIBRATOR == 0)
standard_sn = df[df['IS_CALIBRATOR'] == 0]
plt.scatter(standard_sn['zCMB'], standard_sn['m_b_corr'],
            color='gray', alpha=0.5, s=10, label='Standard Supernovae')

# Plot the calibrators on top (IS_CALIBRATOR == 1)
plt.scatter(calibrators['zCMB'], calibrators['m_b_corr'],
```

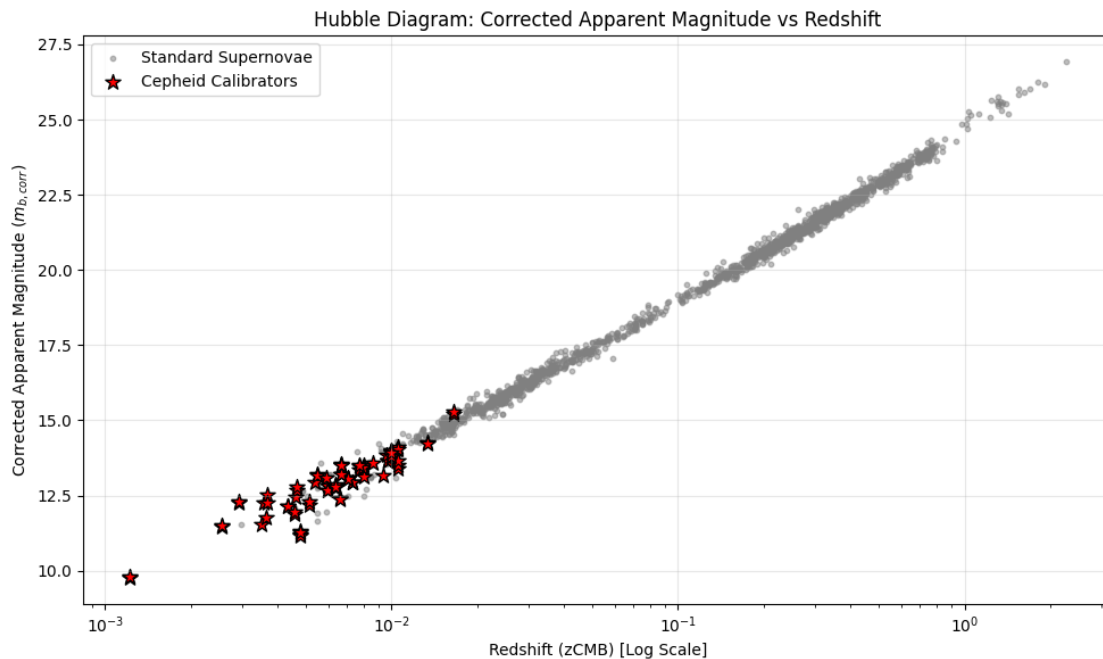
```

        color='red', marker='*', s=100, edgecolor='black', label='Cepheid_
↳Calibrators')

plt.title('Hubble Diagram: Corrected Apparent Magnitude vs Redshift')
plt.xlabel('Redshift (zCMB) [Log Scale]')
plt.ylabel('Corrected Apparent Magnitude ( $m_{b,corr}$ )')

# It is standard practice to plot redshift on a logarithmic scale in Hubble_
↳Diagrams
plt.xscale('log')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```



```

[9]: plt.figure(figsize=(10, 6))

# Plot the standard sample (IS_CALIBRATOR == 0)
standard_sn = df[df['IS_CALIBRATOR'] == 0]
plt.scatter(standard_sn['zCMB'], standard_sn['m_b_corr'],
            color='gray', alpha=0.5, s=10, label='Standard Supernovae')

# Plot the calibrators on top (IS_CALIBRATOR == 1)
plt.scatter(calibrators['zCMB'], calibrators['m_b_corr'],

```

```

        color='red', marker='*', s=100, edgecolor='black', label='Cepheid_
↪Calibrators')

plt.title('Hubble Diagram: Corrected Apparent Magnitude vs Redshift')
plt.xlabel('Redshift (zCMB)')
plt.ylabel('Corrected Apparent Magnitude ( $m_{b,corr}$ )')

plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```

